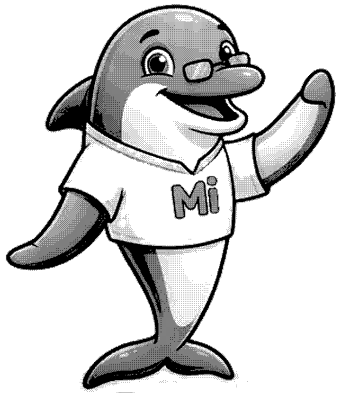
Math Inquirers

# Extension Workbook

## with answers

## Level 4

Self-study | Ages 9-11




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 4**
Self-study practice with answers.


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## Good to know

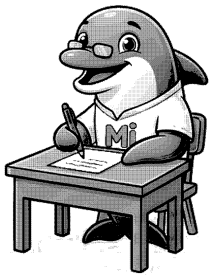
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- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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# Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

## How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

## For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- The rulers in Unit 9 are printed at real size.

## The four page names

- **Start here.** The main path. Do these first.
- **Think harder.** Longer problems. Hints are printed upside down.
- **Go further.** A weekend page. Big challenges, just for fun.
- **Check yourself.** Four quick questions at the end.

# Contents


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| 2 | **Big number puzzles**<br>Not in this proof edition | – |
| 3 | **Operation detectives**<br>Not in this proof edition | – |
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# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | **Factor hunters** | Not in this proof edition | | | | ☐ |
| 2 | **Big number puzzles** | Not in this proof edition | | | | ☐ |
| 3 | **Operation detectives** | Not in this proof edition | | | | ☐ |
| 4 | **Fraction frontier** | Not in this proof edition | | | | ☐ |
| 5 | **Decimal detectives** | Not in this proof edition | | | | ☐ |
| 6 | **Angle turns** | Not in this proof edition | | | | ☐ |
| 7 | **Area architects II** | Not in this proof edition | | | | ☐ |
| 8 | **Symmetry and nets** | Not in this proof edition | | | | ☐ |
| 9 | **Data Detectives** | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 10 | **Puzzle day** | Not in this proof edition | | | | ☐ |
| 11 | **Catch-up week** | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

### Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

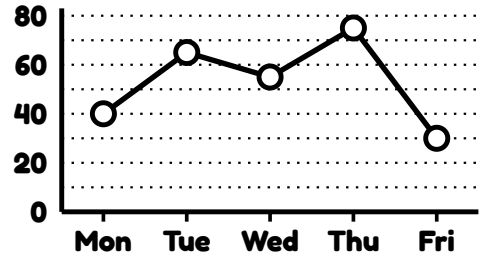
# Learn it: Tables and line graphs

**My goal.** I can read values and changes on line graphs. I can find missing numbers in tables and spot graphs with scales that mislead.

On a line graph, read the title and the scale first. The change between 2 neighbouring points is the difference of their values. A table can have a missing number: use the totals to find it.

## Worked example 1

**The graph shows the cups of barley water a stall sold. Each line is worth 10. How many cups were sold on Wednesday? When was the biggest drop?**




1. The Wednesday point is halfway between 50 and 60: 55 cups.
2. Find each change. Thu to Fri is 75 – 30 = 45 down. That is the biggest drop.

**Answer: 55 cups. The biggest drop was Thursday to Friday.**

## Worked example 2

**Find the missing number. Each row and each column adds to its total.**

| | Week 1 | Week 2 | Total |
|---|---|---|---|
| Ali | 14 | ? | 31 |
| Mei | 20 | 18 | 38 |
| Total | 34 | 35 | 69 |

1. Use Ali's row: 14 + ? = 31, so ? = 31 – 14 = 17.
2. Check the Week 2 column: 17 + 18 = 35.

**Answer: The missing number is 17.**

## How to read data

1. Read the title and the scale. What is one line worth?
2. Change = the difference between 2 neighbours.
3. Missing numbers: use the row or column total.

## Good to know

- The steepest line is the biggest change.
- Check where each scale starts.

## Your turn

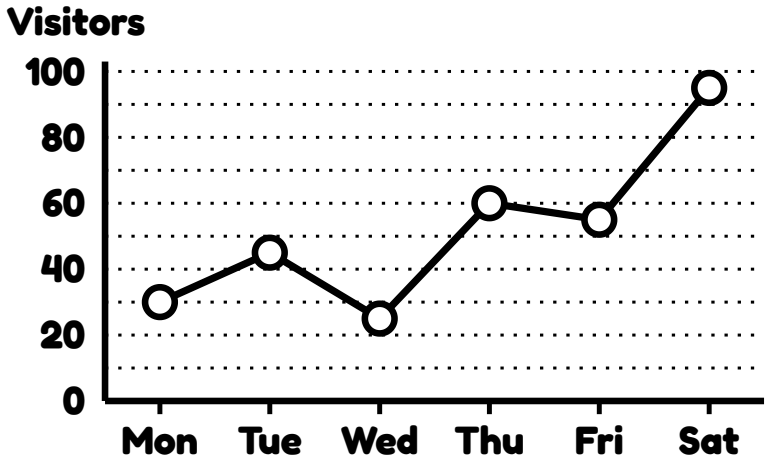
Answer at the back of the book.

Try **A stall sold 35 cups on Monday and 60 cups on Tuesday. By how many cups did sales go up?** ☐

## Start here

Do these first. Take your time.

**1** **The graph shows how many people visited a library each day. Each line is worth 10.**




**(a)** Visitors on Thursday: ☐ **(b)** Visitors on Tuesday: ☐

**(c)** Visitors on Monday and Tuesday together: ☐

**2** **Use the graph in Question 1. Find the biggest increase and the biggest decrease between 2 days.**

**(a)** Biggest increase: from ☐ to ☐, up by ☐

**(b)** Biggest decrease: from ☐ to ☐, down by ☐

Changes: _____

**3** **The table shows the library books 3 classes borrowed. Each row and each column adds up to its total. Write the missing numbers.**

**(a)** Fill in the 6 empty boxes. Go row by row.

**(b)** Which class borrowed more in Week 1 than in Week 2?

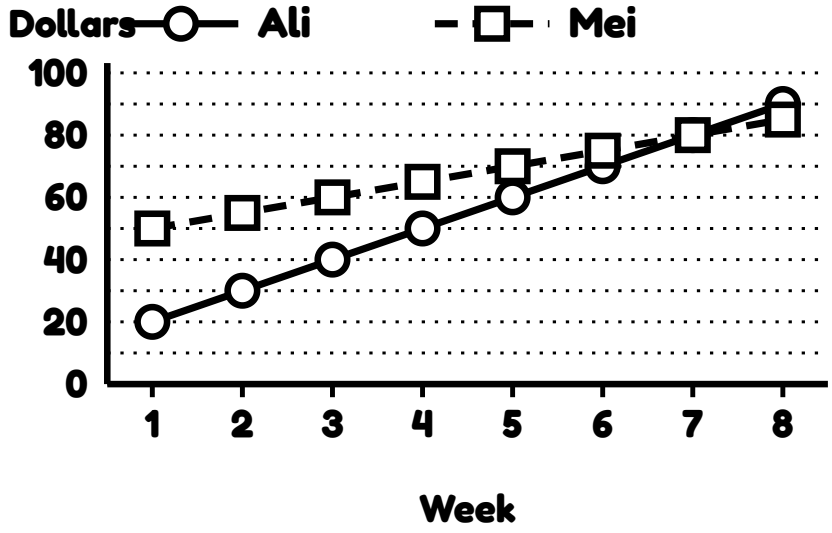
4A 4B 4C

| | Week 1 | Week 2 | Total |
|---|---|---|---|
| 4A | 38 | | 85 |
| 4B | | 29 | 70 |
| 4C | 33 | 41 | |
| Total | | | |

## Start here

Keep going. Check your work.

**4** **The graph shows the money in Ali's and Mei's money boxes at the end of each week.**




**(a)** Mei's money in Week 3: $ ☐ **(b)** Ali saves $ ☐ each week.

**(c)** They have the same amount in Week ☐.

**(d)** Ali first has more than Mei in Week ☐.

**5** **Siti reads more pages each day. She reads the same number more each day. Fill in the table.**

| Day | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Pages | 12 | | | 27 | |

**(a)** She reads ☐ more pages each day.

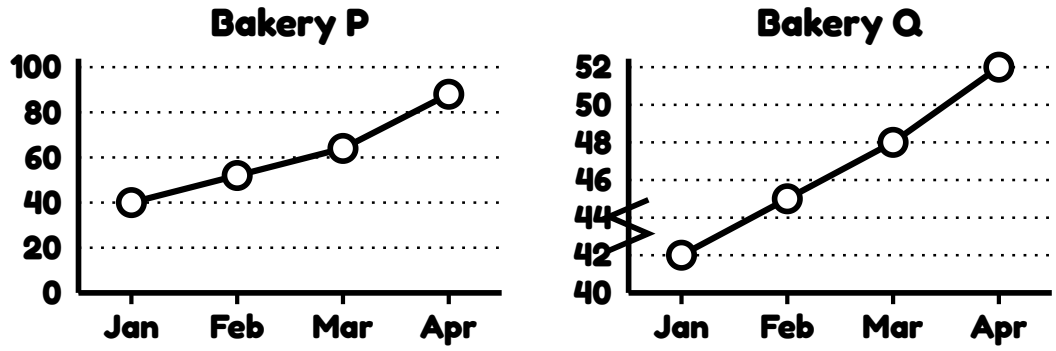
**(b)** In the 5 days she reads ☐ pages.

Working: _____

## Start here

Nearly there. You can do it.

**6** **Two bakeries show the cakes they sold. Look at each scale. Bakery Q's scale starts at 40.**




**(a)** From January to April, Bakery P went up by ☐ cakes.

**(b)** Bakery Q went up by ☐ cakes.

**(c)** Which bakery sold more extra cakes? P Q

**7** **A graph's scale starts at 50. Each line is worth 20. Ali's point is 4 lines above the start. Mei's point is 9 lines above the start.**

**(a)** Ali's number is ☐

**(b)** Mei's number is ☐

**(c)** Mei says: "9 is more than double 4, so my number is more than double Ali's."
Is she right? Yes No

Working: _____

**8** **At 7 am the temperature was 25 °C. It rose by 2 °C every hour until noon. Then it fell by 3 °C every hour.**

**(a)** The temperature at noon was ☐ °C

**(b)** The temperature at 3 pm was ☐ °C

**(c)** At what time was it 29 °C? ☐ am

Working: _____

## Think harder

Hints are upside down. Try first!

**1** **A shop's sales go up by the same amount every week. In Week 3 it sold 52 toys. In Week 8 it sold 87 toys.**

**(a)** It sold ☐ toys in Week 1.

**(b)** It first sold more than 100 toys in Week ☐.

Working: _____

_____

**Hint.** From Week 3 to Week 8 is 5 weeks. How much is that each week?

**2** **A stall sold 200 plates in 4 days. On Day 2 it sold twice as many as on Day 1. On Day 3 it sold 15 more than on Day 2. On Day 4 it sold 25. How many plates did it sell on Day 1?** ☐

Draw a bar model here.

_____

_____

**Hint.** Day 1 is 1 unit. Day 2 is 2 units. Day 3 is 2 units and 15 more.

**3** **Tank A has 100 litres of water and loses 4 litres every hour. Tank B has 15 litres and gains 13 litres every hour. On a line graph the 2 lines cross.**

**(a)** After ☐ hours the tanks hold the same amount.

**(b)** Each tank then holds ☐ litres.

Working: _____

_____

**Hint.** How big is the gap at the start? How much does it close each hour?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Two graphs show the same shop's sales in January and February. On graph X the scale starts at 0 and each line is worth 20. On graph Y the scale starts at 60 and each line is worth 5. Graph X shows January 4 lines up and February 5 lines up.**

**(a)** The sales rose by ☐

**(b)** On graph Y, January is ☐ lines up and February is ☐ lines up.

**(c)** Which graph makes the rise look bigger? X Y

Working: _____

**Hint.** Find the sales in January and February first.

**2** **A stall sold 340 plates from Monday to Friday. Each day it sold 8 more plates than the day before.**

**(a)** On Monday it sold ☐ plates.

**(b)** On Friday it sold ☐ plates.

Working: _____

_____

**Hint.** Take away the extras: 0, 8, 16, 24 and 32.

**3** **Beyond Level 4: Average**
**The average of some numbers is their total divided by how many numbers there are.**

**(a)** The average of 3, 6 and 9 is ☐

**(b)** A stall's average over 4 days is 45 plates. Three days were 40, 50 and 38 plates. The fourth day was ☐ plates.

Working: _____

_____

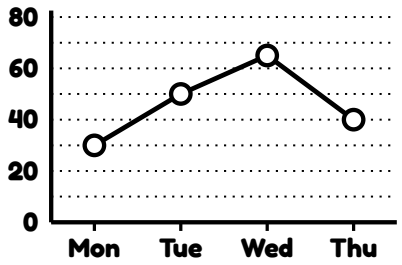
**Hint.** Total = average × the number of days.

## ✔ Check yourself

No hints. Try each one on your own.

**1**

**The graph shows the cakes a shop sold. Each line is worth 10. How many cakes were sold on Wednesday?** ☐




**2** **A shop sold 28 pens on Monday and 35 on Tuesday. The 3-day total was 99.**

**Wednesday:** ☐ **pens**

Working: _____

**3** **Sales rise by the same amount each day. Day 1: 14. Day 4: 32. Day 6:** ☐

Working: _____

**4** **Both graphs show sales of 60 in May and 70 in June. A starts at 0 and B starts at 50. Which looks steeper?** **A** **B**

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

**Tick each box when you finish. Write your Check yourself score.**

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| **1** | Factor hunters | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **2** | Big number puzzles | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **3** | Operation detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **4** | Fraction frontier | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **5** | Decimal detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **6** | Angle turns | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **7** | Area architects II | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **8** | Symmetry and nets | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **9** | Data Detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| **10** | Puzzle day | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

**You finished the Math Inquirers Extension Workbook, Level 4.**

Date

Signed by a grown-up

# Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

# Unit 9 · Data Detectives

**Your turn**
page 6

**Answer:** 25
**Why:** 60 − 35 = 25, so sales went up by 25 cups.

**START HERE**

**1**
page 7

**Answer: (a)** 60 **(b)** 45 **(c)** 75
**Why:** (a) The Thursday point is on the 60 line. (b) The Tuesday point is halfway between 40 and 50: 45. (c) 30 + 45 = 75.

**2**
page 7

**Answer: (a) Friday to Saturday, up by 40 (b) Tuesday to Wednesday, down by 20**
**Why:** The changes are +15, −20, +35, −5 and +40. The biggest increase is Friday to Saturday: 95 − 55 = 40. The biggest decrease is Tuesday to Wednesday: 45 − 25 = 20. The steepest line going up and the steepest line going down show these.

**3**
page 7

**Answer: (a)** 47, 41, 74, 112, 117, 229 **(b)** 4B
**Why:** 4A: 85 − 38 = 47. 4B: 70 − 29 = 41. 4C: 33 + 41 = 74. Week 1: 38 + 41 + 33 = 112. Week 2: 47 + 29 + 41 = 117. Total: 112 + 117 = 229, and 85 + 70 + 74 = 229 too. 4B borrowed 41 in Week 1 and 29 in Week 2.

**4**
page 8

**Answer: (a)** 60 **(b)** 10 **(c)** 7 **(d)** 8
**Why:** (a) The Mei line (dashed, square points) is on 60 in Week 3. (b) Ali's line goes up by 10 each week. (c) The lines meet in Week 7, where both have $80. (d) Ali has $90 in Week 8 and Mei has $85, so Week 8 is the first week Ali has more.

**5**
page 8

**Answer: (a)** 5 **(b)** 110
**Why:** Day 1 to Day 4 is 3 days, and 27 − 12 = 15, so she reads 15 ÷ 3 = 5 more each day. The days are 12, 17, 22, 27 and 32. The total is 12 + 17 + 22 + 27 + 32 = 110 pages.

**6**
page 9

**Answer: (a)** 48 **(b)** 10 **(c)** P
**Why:** (a) P: 88 − 40 = 48. (b) Q: 52 − 42 = 10. (c) P went up by 48 and Q by 10. Bakery Q's line looks steeper only because its scale starts at 40 and each line is worth 2.

**7**
page 9

**Answer: (a)** 130 **(b)** 230 **(c)** No
**Why:** (a) 50 + 4 × 20 = 130. (b) 50 + 9 × 20 = 230. (c) Double Ali's number is 260, and Mei's 230 is less. The lines were counted from 50, not from 0, so counting lines gives the wrong comparison.

**8**
page 9

**Answer: (a)** 35 **(b)** 26 **(c)** 9
**Why:** (a) From 7 am to noon is 5 hours: 25 + 5 × 2 = 35 °C. (b) From noon to 3 pm is 3 hours: 35 − 3 × 3 = 26 °C. (c) 29 − 25 = 4 °C more, and 4 ÷ 2 = 2 hours after 7 am, which is 9 am.

## THINK HARDER

**1** page 10

**Answer: (a) 38 (b) 10**
**Why:** 87 − 52 = 35 over 5 weeks, so sales go up by 7 each week. Week 1 is 2 weeks before Week 3: 52 − 14 = 38. Week 9 is 94 and Week 10 is 101, so Week 10 is the first week above 100.

**2** page 10

**Answer: 32**
**Why:** Take away Day 4 and the extra 15: 200 − 25 − 15 = 160. Day 1 is 1 unit, Day 2 is 2 units and Day 3 is 2 units: 5 units = 160. 1 unit = 32. Check: 32 + 64 + 79 + 25 = 200.

**3** page 10

**Answer: (a) 5 (b) 80**
**Why:** At the start the gap is 100 − 15 = 85 litres. Each hour it closes by 4 + 13 = 17 litres. 85 ÷ 17 = 5 hours. Tank A: 100 − 5 × 4 = 80 litres. Tank B: 15 + 5 × 13 = 80 litres.

## GO FURTHER

**1** page 11

**Answer: (a) 20 (b) 4 lines and 8 lines (c) Y**
**Why:** On X, January is 4 × 20 = 80 and February is 5 × 20 = 100, so sales rose by 20. On Y: (80 − 60) ÷ 5 = 4 lines and (100 − 60) ÷ 5 = 8 lines. Graph Y's line goes up 4 lines and X's goes up 1 line, so Y makes the rise look bigger.

**2** page 11

**Answer: (a) 52 (b) 84**
**Why:** The extras over Monday are 0 + 8 + 16 + 24 + 32 = 80. Without them, 5 equal days make 340 − 80 = 260, so Monday is 260 ÷ 5 = 52. Friday is 52 + 32 = 84. The days are 52, 60, 68, 76 and 84.

**3** page 11

**Answer: (a) 6 (b) 52**
**Why:** (a) 3 + 6 + 9 = 18, and 18 ÷ 3 = 6. (b) The total for 4 days is 4 × 45 = 180. The 3 known days make 40 + 50 + 38 = 128. The fourth day is 180 − 128 = 52 plates.

## CHECK YOURSELF

**1** page 12

**Answer: 65**
**Why:** The Wednesday point is halfway between the 60 line and the 70 line: 65 cakes.

**2** page 12

**Answer: 36**
**Why:** 28 + 35 = 63 pens on Monday and Tuesday. 99 − 63 = 36 pens on Wednesday.

**3** page 12

**Answer: 44**
**Why:** From Day 1 to Day 4 is 3 days, and 32 − 14 = 18, so sales go up by 6 each day. Day 6 is 32 + 2 × 6 = 44.

**4** page 12

**Answer: B**
**Why:** Graph B's scale starts at 50, so the same 10 more looks like a bigger rise. Sales rose by 10 on both graphs.